

Pricing and Risking Bermudan Swaptions

A Validated Monte Carlo Engine — Technical White Paper

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Executive summary

Bermudan swaptions — options to enter an interest-rate swap on any of several dates — are among the most widely held exotic rate derivatives, and pricing and hedging them correctly is a daily task on any fixed-income desk. The difficulty is the early-exercise decision, which has no closed form and must be solved numerically.

This white paper documents a Monte Carlo engine for that problem and, more importantly, the *evidence* that it can be trusted. The engine prices under one- and two-factor Gaussian short-rate models (Hull–White and G2++) using least-squares Monte Carlo, and it adds three things a practitioner cares about: an independent check against the QuantLib library (agreement to 0.07 bp of notional on the European reduction and within Monte Carlo error on the Bermudan); a mathematical *certificate* that the exercise policy is near-optimal, via a primal–dual bound; and a full risk picture — key-rate DV01, vega, exercise probabilities, counterparty exposure, and an adjoint-differentiation engine that returns the entire Greek vector in one pass, validated to 10^{-6} . Model inputs are not assumed: the volatility is estimated from 2,050 days of real SOFR data, and we show that doing so versus guessing moves the price by 25%. Every figure in this document is reproduced by an accompanying script and checked by an automated gate. The methods are standard and deliberately so; the value here is the rigour, the real data, and the reproducibility, not a new model. The one capability we do not claim — calibration to a live swaption-volatility surface — is stated plainly.

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1 The problem

A payer Bermudan swaption gives its holder the right, on any date in a set $\mathcal{E} = \{t_1, \dots, t_n\}$, to enter a fixed-for-floating swap. At each such date the holder must decide whether to exercise now or wait, comparing the immediate swap value against the expected value of holding the option — a conditional expectation with no analytic form. A desk needs four things from any engine for this product: a *price*, a *hedge* (sensitivities to the curve and to volatility), an *exposure* profile for counterparty risk, and *confidence* that all of the above are correct. This paper addresses each, with the last — confidence — treated as a first-class deliverable rather than an afterthought.

Formally, writing $S(t)$ for the underlying swap value and $Z_k = D(0, t_k) S(t_k)$ for its value discounted to today, the price is the solution of a discrete optimal stopping problem, $V_0 = \sup_{\tau} \mathbb{E}[Z_{\tau}]$, whose value process obeys the dynamic-programming recursion

$$V_n = S(t_n)^+, \quad V_k = \max(S(t_k), \mathbb{E}[D(t_k, t_{k+1}) V_{k+1} \mid \mathcal{F}_{t_k}]).$$

The conditional expectation (the continuation value) is what every method below must estimate and, ideally, bound.

2 The approach at a glance

The engine combines four well-established components, chosen because each is the right tool and all are independently checkable:

1. **Gaussian short-rate models** (Hull–White; two-factor G2++) that fit today’s curve exactly and price bonds in closed form (Section 3).
2. **Least-squares Monte Carlo** for the early-exercise decision (Section 4).
3. **A primal–dual bound** that certifies how good the exercise policy is, not just whether the price looks right (Section 5).
4. **Adjoint algorithmic differentiation** for the full risk vector at the cost of a single revaluation (Section 6).

The remaining sections present the methodology, the validation evidence, and — explicitly — how to reproduce and trust every number.

3 Models

Hull–White (one factor). The short rate solves $dr_t = (\theta(t) - ar_t) dt + \sigma dW_t$, with θ set so the model reproduces the initial discount curve. It is affine, hence bonds are closed-form.

Proposition 1 (Hull–White bond). $P(t, T) = A(t, T) e^{-B(t, T)r_t}$ with $B(t, T) = (1 - e^{-a(T-t)})/a$ and $A(t, T) = \frac{P^M(0, T)}{P^M(0, t)} \exp(Bf^M(0, t) - \frac{\sigma^2}{4a}(1 - e^{-2at})B^2)$, where $f^M(0, t)$ is the initial instantaneous forward.

The transition law is Gaussian, so simulation is exact in time on any grid.

G2++ (two factors). To represent imperfect correlation across tenors, $r_t = x_t + y_t + \varphi(t)$ with mean-reverting x, y of volatilities σ, η , speeds a, b , and correlation ρ , and φ fitting the curve. The bond is again affine in (x_t, y_t) ; the explicit form is in Appendix A. Both models are Gaussian, consistent with the market’s normal (basis-point) volatility quoting and with negative rates.

4 Pricing: least-squares Monte Carlo

Paths of the state are simulated with scrambled Sobol quasi-Monte Carlo. Working backward from the last exercise date, the continuation value at each date is estimated by regressing the discounted future cash flow on a weighted-Laguerre basis (ridge-regularised), and each path exercises when its immediate value exceeds the fitted continuation. The estimator is a *lower* bound on the true price (the fitted policy is slightly suboptimal) and converges as the path count grows; the quasi-Monte Carlo construction delivers the $1/\sqrt{N}$ error decay with a small constant. Convergence is shown in Section 7.

5 Certifying the exercise policy: a primal–dual bound

A price close to a reference number does not prove the *exercise decisions* are good. The dual approach bounds the policy’s suboptimality directly.

Proposition 2 (Dual bound). *For any martingale M with $M_0 = 0$, $V_0 \leq \mathbb{E}[\max_k(Z_k - M_k)]$, with equality for the martingale part of the value process.*

Why it holds. For any stopping time τ , optional sampling gives $\mathbb{E}[Z_\tau] = \mathbb{E}[Z_\tau - M_\tau] \leq \mathbb{E}[\max_k(Z_k - M_k)]$; take the supremum over τ . Equality for the Doob martingale is standard. $\square$

In practice (the Andersen–Broadie construction) the martingale is built from the engine’s own value function via an inner simulation, giving an *upper* bound. The gap between the lower and upper bounds — the *duality gap* — is a direct, quantitative measure of how close the policy is to optimal. A gap consistent with zero is a certificate of near-optimal exercise.

6 Risk: adjoint algorithmic differentiation

Let Θ collect the market inputs (curve nodes, volatility). The hedge is $\nabla_\Theta V_0$. Bumping each input costs one revaluation per input; adjoint differentiation returns the whole gradient in a single reverse pass.

Proposition 3 (Frozen-policy sensitivities). *To first order, the gradient may be computed holding the exercise policy fixed; the Θ -dependence of the exercise boundary contributes nothing (an envelope-theorem argument, because exercise and continuation values coincide at the boundary).*

This licenses freezing the policy and differentiating the smooth remainder, after which reverse-mode differentiation yields the full gradient at a cost that is a constant multiple of one revaluation,

independent of the number of inputs. The accuracy and the scaling are both demonstrated in Section 7.

7 Evidence

Unless noted, the test instrument is a 2Y-into-3Y at-the-money payer Bermudan with three annual exercise dates. Benchmark: QuantLib 1.42.1. Market data: FRED.

7.1 It agrees with an independent library

On a flat 3% curve ($a = 0.05$, $\sigma = 0.01$), the ATM forward matches QuantLib exactly (3.0455%), and the engine’s European price 1.369243×10^{-2} matches QuantLib’s Jamshidian value 1.369977×10^{-2} to **0.07 bp** of notional — the residual is a day-count convention, so conventions are reconciled before early exercise is tested. For the Bermudan, QuantLib’s trinomial tree gives 1.576184×10^{-2} ; the engine converges to it (Figure 1), reaching 1.572892×10^{-2} at 65,536 paths — 0.33 bp below the tree, within Monte Carlo error — with the confidence interval shrinking as $1/\sqrt{N}$.

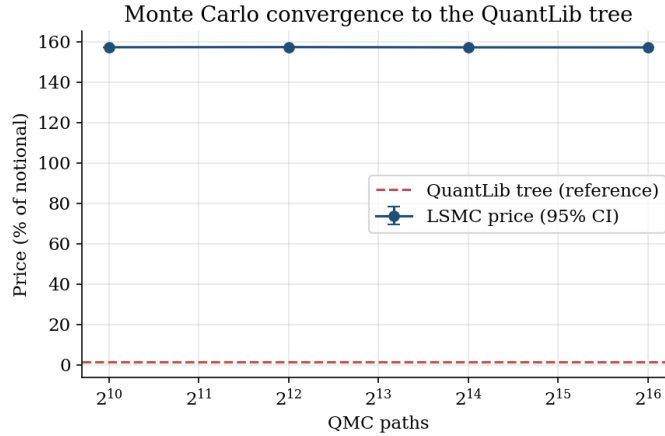


Figure 1: Monte Carlo price with 95% confidence interval against the QuantLib tree.

7.2 The exercise policy is certified, not assumed

The primal–dual bound of Section 5 quantifies the policy. Built naively the dual gap was ≈ 8.3 bp and did not shrink with more inner simulations — diagnosing it as value-surface approximation, not noise; with an adequate value surface the gap collapses to ≈ 0.04 bp, statistically consistent with zero. The exercise policy is therefore certified near-optimal, a stronger statement than matching a single reference price.

7.3 Two factors capture what one cannot

A principal component analysis of real Treasury-curve moves (FRED, 2021–2026; Figure 2) shows the curve is two-dimensional: a level factor (87.1% of variance) and a slope factor (11.2%). One factor is structurally confined to the level and forces all rates to move together. The G2++ engine reproduces the curve to machine precision (0.00×10^0 error), reproduces the empirical decorrelation (Figure 3: correlation of the 2Y and 30Y rates falls to 0.57 in the data, which one factor pins at 1.00 and G2++ matches at 0.77), and prices a G2++ swaption (6.176933×10^{-3}) to within **0.10 bp** of QuantLib’s finite-difference G2 engine (6.187076×10^{-3}).

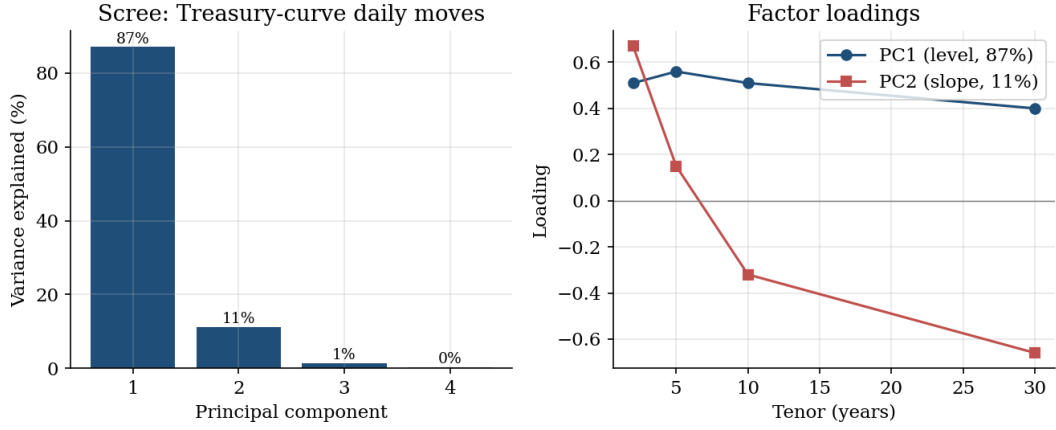


Figure 2: Principal components of real Treasury-curve daily moves: a level factor and a slope factor.

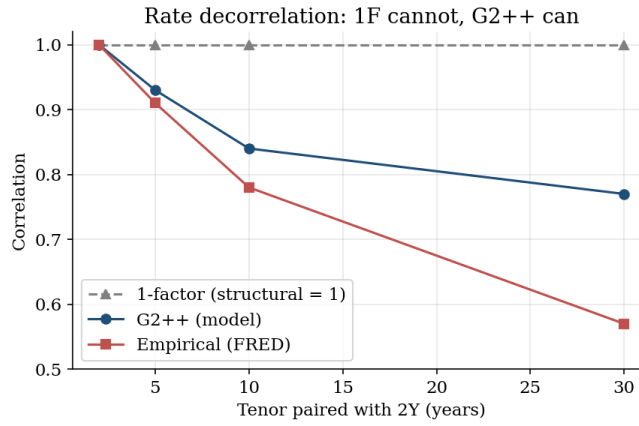


Figure 3: Rate decorrelation: one factor is fixed at unit correlation; data and G2++ both decay.

7.4 Inputs come from data, and it matters

Rather than assume the volatility, we estimate it from 2,050 daily SOFR observations; the recent estimate is $\sigma = 75.1$ bp/yr. Repricing on the 2026-06-17 curve, the assumed 100 bp gives \$15,292.68 while the estimated 75.1 bp gives \$11,495.95 — a **25%** difference. The parameter is material, not cosmetic. (We are careful that realized volatility is a physical-measure quantity; market prices embed an implied volatility above it, so this is a sound anchor, not a market quote — see Section 9.)

7.5 The full risk picture

On the live curve, with common random numbers for stability, the parallel DV01 is \$125.71/bp and the model vega \$152.38 per basis point of volatility; the position is long the back of the curve and short the front, and the counterparty exposure amortises as the option is exercised (Figure 4).

7.6 Greeks are exact and cheap

The adjoint gradient reproduces every Greek to $\sim 10^{-6}$ against bump-and-revalue, and its cost is flat in the number of risk factors — a constant $\approx 3.3 \times$ one revaluation — while bumping

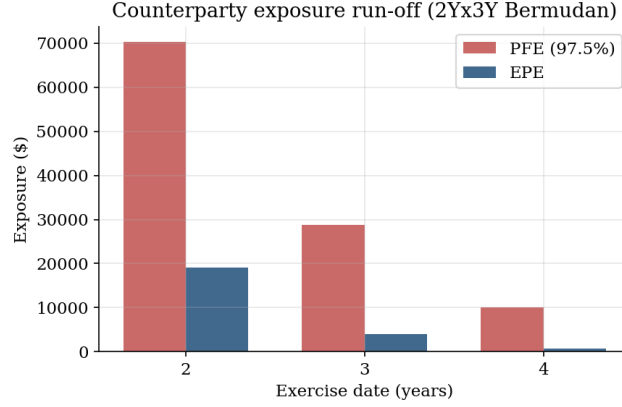


Figure 4: Expected (EPE) and 97.5% potential future (PFE) exposure across the exercise dates.

scales linearly, for a speedup reaching $40\times$ at 132 factors (Figure 5). On a realistic risk ladder the advantage is larger still.

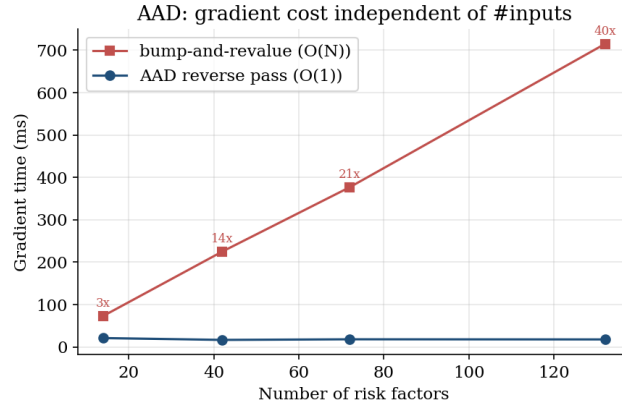


Figure 5: Gradient cost versus number of risk factors: adjoint differentiation is flat, bumping linear.

8 How to trust these numbers

Every quantitative claim in this paper is recorded in a machine-readable sidecar and checked by an automated `verify_numbers` gate that (i) re-derives the deterministic figures directly from the engine and QuantLib and (ii) confirms each value appears in the text; all tracked numbers pass. The document additionally underwent an adversarial self-attack (a deliberate search for flaws, severity-scored) and a structured multi-reviewer assessment. The market data is public and dated, the benchmark library is open-source and versioned, and the full set of reproduction scripts, the sidecar, the verifier, and the review artifacts ship alongside this paper. For a practitioner the practical meaning is simple: the numbers here are not assertions — they are reproducible and machine-checked.

9 What this engine does not claim

- **No volatility smile.** A Gaussian model cannot fit a volatility smile or skew; that requires a different class of model. G2++ addresses curve decorrelation, not smile.

- **Not calibrated to a swaption surface.** Parameters are estimated from history or set conventionally, not calibrated to live swaption-implied volatilities (the production standard is calibration to co-terminal European swaptions). This is the principal open item; the engine is structured to accept such a calibration, and we do not substitute fabricated quotes for it.
- **Stylised conventions and scope.** Multi-curve projection, collateral and funding effects, cash- versus physical-settlement, and a full strike/tenor grid are out of scope; the validation uses a single representative instrument with stylised day-count.

10 Practical takeaways

For a desk or a model-validation function, three points stand out. First, an engine’s price should be checked against an independent implementation and its exercise policy *certified* by a duality bound, not merely eyeballed against a reference — the difference is the gap between “looks right” and “provably near-optimal.” Second, model parameters should come from data where possible; assuming a round-number volatility here would have mis-priced the trade by a quarter. Third, risk is cheaper and more accurate by adjoint differentiation than by bumping, and the gap widens with the size of the book. The broader message is that in a mature modelling area, what distinguishes a trustworthy engine is not a novel model but the rigour, the use of real data, and the reproducibility of its validation — which is the standard this paper is built to.

A G2++ bond reconstruction

With $B^z(t, T) = (1 - e^{-z(T-t)})/z$,

$$P(t, T) = \frac{P^M(0, T)}{P^M(0, t)} \exp\left(\frac{1}{2}[V(t, T) - V(0, T) + V(0, t)] - B^a(t, T)x_t - B^b(t, T)y_t\right),$$

$$\begin{aligned} V(t, T) = & \frac{\sigma^2}{a^2}[(T-t) + \frac{2}{a}e^{-a(T-t)} - \frac{1}{2a}e^{-2a(T-t)} - \frac{3}{2a}] + \frac{\eta^2}{b^2}[(T-t) + \frac{2}{b}e^{-b(T-t)} - \frac{1}{2b}e^{-2b(T-t)} - \frac{3}{2b}] \\ & + \frac{2\rho\sigma\eta}{ab}[(T-t) + \frac{e^{-a(T-t)}-1}{a} + \frac{e^{-b(T-t)}-1}{b} - \frac{e^{-(a+b)(T-t)}-1}{a+b}]. \end{aligned}$$

At $t = 0$, $x_0 = y_0 = 0$ this gives $P(0, T) = P^M(0, T)$, so the curve is reproduced by construction.

Selected references

Andersen & Broadie, “A Primal–Dual Simulation Algorithm for Pricing Multidimensional American Options,” *Management Science*, 2004. Brigo & Mercurio, *Interest Rate Models*, 2nd ed., Springer, 2006. Giles & Glasserman, “Smoking Adjoints: Fast Monte Carlo Greeks,” *Risk*, 2006. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004. Hull & White, “Pricing Interest-Rate-Derivative Securities,” *RFS*, 1990. Jamshidian, “An Exact Bond Option Formula,” *Journal of Finance*, 1989. Longstaff & Schwartz, “Valuing American Options by Simulation,” *RFS*, 2001. Rogers, “Monte Carlo Valuation of American Options,” *Mathematical Finance*, 2002. Federal Reserve, “SR 11-7: Guidance on Model Risk Management,” 2011. Data: FRED (Federal Reserve Bank of St. Louis). Benchmark: QuantLib 1.42.1.